

eCommerce Customer Ranking Using BigQuery ML

An End-to-End Machine Learning Case Study

AUTHOR: Vishal Mahajan

Executive Summary

This MVP delivers a production-grade machine learning solution for predicting customer purchase propensity using Google BigQuery ML, eliminating the need for any external ML infrastructure. Leveraging millions of Google Analytics interaction records from the Google Merchandise Store—natively stored and processed in BigQuery—the project trains a logistic regression model directly through SQL to analyze browsing behavior, engagement patterns, and historical purchase signals at scale.

Rather than relying on binary classification, the solution reframes the problem as a Top-K propensity ranking task, enabling marketers to identify the highest-likelihood customers for targeted outreach. This cloud-native approach provides accurate, scalable, and cost-efficient predictive analytics while operating entirely within the BigQuery ecosystem. The resulting model serves as a strong baseline for real-world ecommerce personalization, customer segmentation, and ROI-driven marketing activation.

Problem Statement

The core challenge is to transform millions of raw Google Analytics records from the Google Merchandise Store into actionable intelligence that improves marketing efficiency. Traditional binary classification approaches struggle with extreme class imbalance and fail to reliably identify customers who are most likely to return and complete a purchase. To address this, the project reframes the task as a propensity-ranking problem: using BigQuery ML to score each customer by their likelihood of purchasing on a return visit and selecting the top-K highest-propensity customers for targeted marketing outreach. This enables data-driven personalization, smarter budget allocation, and measurable lift in conversion outcomes.

Dataset

Attribute	Detail
Source	eCommerce Web Analytics data via BigQuery Public Datasets
Time Range	2016 – 2017
Total Records	3.3 M+
Training Set	2.5 M+
Test Set	470 K
Target Variable	will_buy_on_return_visit

Feature Engineering

Features were chosen based on domain knowledge and correlation analysis:

- latest_ecommerce_progress -- based on action_type (click through list, details, add to cart, remove from cart, checkout, complete purchase, refund, etc.)
- bounces -- Total bounces. For a bounced session, the value is 1, otherwise it is null
- Time_on_site -- Total time of the session expressed in seconds.
- pageviews -- Total number of pageviews within the session.
- source (traffic source) -- Traffic Source from which the session originated.
- medium (traffic medium) -- The medium of the traffic source (Organic, CPC, Referral).
- channelGrouping -- The Default Channel Group associated with an end user's session.

- deviceCategory -- The type of device (Mobile, Tablet, Desktop)
- country -- The country from which sessions originated, based on IP address

Data filters: newVisist equals 1.

Architecture & ML Workflow

01	Data Ingestion	eCommerce Web Analytics data via BigQuery Public Datasets
02	Data Cleaning	SQL WHERE filters records for only new visits
03	Feature Creation	MAX (action_type) per session computed inline in SQL SELECT
04	Model Training	CREATE OR REPLACE MODEL with BQML LOGISTIC_REG on training rows
05	Evaluation	ML.EVALUATE returns ROC_AUC. For ranking use case PRECISION, RECALL, ACCURACY, R1 score ignored.
06	Prediction	ML.PREDICT scores each customer by their likelihood of purchasing on a return visit

Key SQL

Model training executed entirely in BigQuery SQL -- no Python or external tooling required

```
CREATE OR REPLACE MODEL `ecommerce.classification_model`
OPTIONS
  (model_type='logistic_reg', input_label_cols = ['will_buy_on_return_visit']) AS
WITH all_visitor_stats AS (
SELECT
  fullvisitorid,
  IF(COUNTIF(totals.transactions > 0 AND totals.newVisits IS NULL) > 0, 1, 0) AS will_buy_on_return_visit
FROM `data-to-insights.ecommerce.web_analytics`
GROUP BY fullvisitorid
)
```

```
# add in new features
SELECT * EXCEPT(unique_session_id) FROM (
  SELECT
    CONCAT(fullvisitorid, CAST(visitId AS STRING)) AS unique_session_id,
    # input_label_cols
    will_buy_on_return_visit,
    MAX(CAST(h.eCommerceAction.action_type AS INT64)) AS latest_ecommerce_progress,
    # behavior on the site
    IFNULL(totals.bounces, 0) AS bounces,
    IFNULL(totals.timeOnSite, 0) AS time_on_site,
    IFNULL(totals.pageviews, 0) AS pageviews,
    # where the visitor came from
    trafficSource.source,
    trafficSource.medium,
    channelGrouping,
    # mobile or desktop
    device.deviceCategory,
    # geographic
    IFNULL(geoNetwork.country, '') AS country
  FROM `data-to-insights.ecommerce.web_analytics`,
  UNNEST(hits) AS h
  JOIN all_visitor_stats USING(fullvisitorid)
  WHERE 1=1
  # only predict for new visits
  AND totals.newVisits = 1
  AND date BETWEEN '20160801' AND '20170430' # train 9 months
  GROUP BY
```

```

unique_session_id,
will_buy_on_return_visit,
bounces,
time_on_site,
totals.pageviews,
trafficSource.source,
trafficSource.medium,
channelGrouping,
device.deviceCategory,
country
);

```

Model Evaluation

```

SELECT
roc_auc,
CASE
  WHEN roc_auc > .9 THEN 'good'
  WHEN roc_auc > .8 THEN 'fair'
  WHEN roc_auc > .7 THEN 'decent'
  WHEN roc_auc > .6 THEN 'not great'
  ELSE 'poor' END AS model_quality
FROM
  ML.EVALUATE(MODEL ecommerce.classification_model, (
WITH all_visitor_stats AS (
SELECT
fullvisitorid,
IF(COUNTIF(totals.transactions > 0 AND totals.newVisits IS NULL) > 0, 1, 0) AS will_buy_on_return_visit
FROM `data-to-insights.ecommerce.web_analytics`
GROUP BY fullvisitorid
)

```

add in new features

```

SELECT * EXCEPT(unique_session_id) FROM (
SELECT
  CONCAT(fullvisitorid, CAST(visitId AS STRING)) AS unique_session_id,
  # input_label_cols
  will_buy_on_return_visit,
  MAX(CAST(h.eCommerceAction.action_type AS INT64)) AS latest_ecommerce_progress,
  # behavior on the site
  IFNULL(totals.bounces, 0) AS bounces,
  IFNULL(totals.timeOnSite, 0) AS time_on_site,
  totals.pageviews,
  # where the visitor came from
  trafficSource.source,
  trafficSource.medium,
  channelGrouping,
  # mobile or desktop
  device.deviceCategory,
  # geographic
  IFNULL(geoNetwork.country, '') AS country
FROM `data-to-insights.ecommerce.web_analytics`,
  UNNEST(hits) AS h
JOIN all_visitor_stats USING(fullvisitorid)
WHERE 1=1
  # only predict for new visits
  AND totals.newVisits = 1
  AND date BETWEEN '20170501' AND '20170630' # eval 2 months
GROUP BY
unique_session_id,
will_buy_on_return_visit,
bounces,

```

```

time_on_site,
totals.pageviews,
trafficSource.source,
trafficSource.medium,
channelGrouping,
device.deviceCategory,
country
)
));

```

ROC_AUC provides a reliable indication of who is more likely to buy on subsequent visit.

Model Prediction

```

SELECT
*
FROM
ml.PREDICT(MODEL `ecommerce.classification_model`,
(

WITH all_visitor_stats AS (
SELECT
fullvisitorid,
IF(COUNTIF(totals.transactions > 0 AND totals.newVisits IS NULL) > 0, 1, 0) AS will_buy_on_return_visit
FROM `data-to-insights.ecommerce.web_analytics`
GROUP BY fullvisitorid
)

```

```

SELECT
CONCAT(fullvisitorid, '-',CAST(visitId AS STRING)) AS unique_session_id,
# input_label_cols
will_buy_on_return_visit,
MAX(CAST(h.eCommerceAction.action_type AS INT64)) AS latest_ecommerce_progress,
# behavior on the site
IFNULL(totals.bounces, 0) AS bounces,
IFNULL(totals.timeOnSite, 0) AS time_on_site,
totals.pageviews,
# where the visitor came from
trafficSource.source,
trafficSource.medium,
channelGrouping,
# mobile or desktop
device.deviceCategory,
# geographic
IFNULL(geoNetwork.country, '') AS country
FROM `data-to-insights.ecommerce.web_analytics`,
UNNEST(hits) AS h
JOIN all_visitor_stats USING(fullvisitorid)
WHERE
# only predict for new visits
totals.newVisits = 1
AND date BETWEEN '20170701' AND '20170801' # test 1 month
GROUP BY
unique_session_id,
will_buy_on_return_visit,
bounces,
time_on_site,
totals.pageviews,
trafficSource.source,
trafficSource.medium,
channelGrouping,
device.deviceCategory,
country
))

```

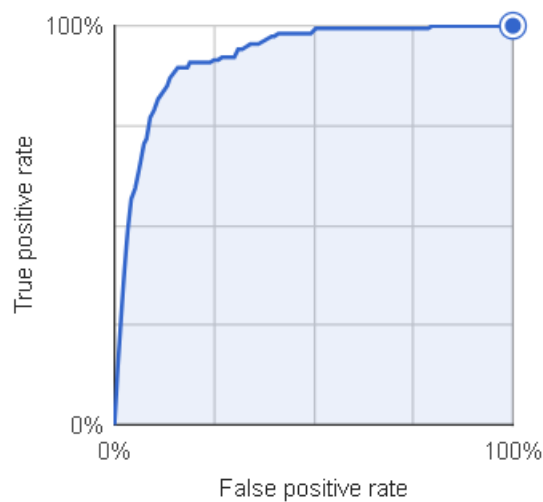
ORDER BY
predicted_will_buy_on_return_visit DESC;

Model Evaluation Results

ML.EVALUATE output on the holdout test set

ROC_AUC	0.92	Separate buyers from non-buyers
ACCURACY	0.97	97 – 99% will not buy. Not useful for the problem statement
PRECISION	0.11	Not useful for the problem statement
RECALL	0.05	Not useful for the problem statement
F1-score	0.07	Not useful for the problem statement

ROC curve 



Area under curve: 0.922

Model Prediction

Google Cloud

qwiklabs-gcp-03-5202f259149c

Search (/) for resources, docs, products, and more

Search

Open project picker (Ctrl O)

Sandbox

Set up billing to upgrade to the full BigQuery experience. [Learn more](#)

Dismiss

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Untitled query

Run

Save

Schedule

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File

Edit

Run

Tools

```
1 SELECT
2 *
3 FROM
4   ml.PREDICT(MODEL `ecommerce.classification_model`,
5   |
6   |
7 WITH all_visitor_stats AS (
8   SELECT
9     fullvisitorid,
10    IF(COUNTIF(totals.transactions > 0 AND totals.newVisits IS NULL) > 0, 1, 0) AS will_buy_on_return_visit
11 FROM `data-to-insights.ecommerce.web_analytics`
12
```

✔ This query will process 120.01 MB when run.

Using on-demand processing quota

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Query results

Chat

View all jobs

Save results

Open in

Job information

Results

Visualization

JSON

Execution details

Execution graph

Row	predicted_will_buy_on_return_visit	predicted_will_bu...	predicted_will_buy_on_retu...	unique_session_id	will_buy_on_retur...	latest_ecommerc...	bounces	time_on_site	pageviews
1	1	1	0.64087120687308741	3598499390128517362-14993...	0	4	0	2347	
		0	0.35912879312691259						
2	1	1	0.55927089463700619	5847392129774736841-15014...	0	6	0	685	
		0	0.44072910536299381						

Results per page: 50 1 - 50 of 59602

Business Insights

- Of the **top 6% of first-time visitors** (sorted in decreasing order of predicted probability), **more than 6%** make a purchase on a later visit.
- Overall, **only 0.6% of first-time visitors make a purchase on a later visit** (based on the actual label “will_buy_on_return_visit”).
- **Targeting** the **top 6%** of first-time **increases marketing ROI** by **9x** vs targeting them all.
- **Zero infrastructure**: BigQuery ML eliminates compute cluster management; training runs as a SQL query.
- **Cost efficiency**: Processing millions of rows in BigQuery is dramatically cheaper than equivalent Spark clusters.
- **Scalability**: The same SQL model could be retrained on data from 2018 - 2026 with a single query modification.

Key Outcomes

- ✓ Built a fully functional ranking model MVP using BigQuery ML (train → evaluate → predict)
- ✓ Used ROC_AUC to evaluate model performance i.e. rank eCommerce customers based on the likelihood of who will purchase on return visit.
- ✓ Ignored PRECISION. RECALL, ACCURACY, F1-SCORE because of class imbalance
- ✓ Model training executed entirely in BigQuery SQL -- Python or external tooling not required.